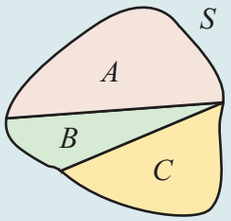
**Solution**

Since the experiment has exactly the three possible mutually exclusive outcomes $A$, $B$ and $C$, they must be exhaustive events.

$$\Rightarrow S = A \cup B \cup C$$




Therefore, by axioms of probability

$P(A) \geq 0, \quad P(B) \geq 0, \quad P(C) \geq 0$ and

$P(A \cup B \cup C) = P(A) + P(B) + P(C) = P(S) = 1$

(i) Given that $P(A) = \frac{4}{7} \geq 0$, $P(B) = \frac{1}{7} \geq 0$, and $P(C) = \frac{2}{7} \geq 0$

Also $P(S) = P(A) + P(B) + P(C) = \frac{4}{7} + \frac{1}{7} + \frac{2}{7} = 1$

Therefore the assignment of probability is permissible.

(ii) Given that $P(A) = \frac{2}{5} \geq 0$, $P(B) = \frac{1}{5} \geq 0$, and $P(C) = \frac{3}{5} \geq 0$

But $P(S) = P(A) + P(B) + P(C) = \frac{2}{5} + \frac{1}{5} + \frac{3}{5} = \frac{6}{5} > 1$

Therefore the assignment is not permissible.

(iii) Since $P(C) = -0.2$ is negative, the assignment is not permissible.

(iv) The assignment is permissible because

$P(A) = \frac{1}{\sqrt{3}} \geq 0$, $P(B) = 1 - \frac{1}{\sqrt{3}} \geq 0$, and $P(C) = 0 \geq 0$

$P(S) = P(A) + P(B) + P(C) = \frac{1}{\sqrt{3}} + 1 - \frac{1}{\sqrt{3}} + 0 = 1.$

(v) Even though $P(A) = 0.421 \geq 0$, $P(B) = 0.527 \geq 0$, and $P(C) = 0.042 \geq 0$, the sum of the probability

$P(S) = P(A) + P(B) + P(C) = 0.421 + 0.527 + 0.042 = 0.990 < 1.$

Therefore, the assignment is not permissible.

**Example 12.2**

An integer is chosen at random from the first ten positive integers. Find the probability that it is (i) an even number (ii) multiple of three.

**Solution**

The sample space is

$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $n(S) = 10$

Let $A$ be the event of choosing an even number and

$B$ be the event of choosing an integer multiple of three.

$A = \{2, 4, 6, 8, 10\}$, $n(A) = 5$,

$B = \{3, 6, 9\}$, $n(B) = 3$



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